Sejal Singh

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SKILLS

Languages: JavaScript, TypeScript, Java, C++, C, Python
Web & Frameworks: React.js, Next.js, Node.js, HTML5, CSS3, Vite, Tailwind CSS, REST APIs
Database: DBMS, MongoDB, MS SQL, PostgreSQL
Tools/Platforms: Git, GitHub, Vercel, Render, Cisco Packet Tracer
Soft Skills: Problem-Solving, Analytical Thinking, Teamwork, Adaptability

PROJECTS

Library Seat Management System | React.js, JavaScript, Vite, Tailwind CSS| Github | Live — Aug' 26

- Architected a **library seat management web application** using **React.js, JavaScript**, **Tailwind CSS** and **Vite** to manage and display **real-time** seat availability across multiple library levels.
- Implemented **interactive seat selection**, level-wise navigation, **seat availability tracking**, search, **filtering**, **reservation functionality**, and **React.js** state management.
- Built reusable **React components** and a responsive **UI** with **HTML5, CSS3**, and **Vite**, focusing on clean **UI/UX** and smooth interactions across devices.

Knight's Tour Problem Game| C++, JavaScript, Node.js | Github | Live — May' 26

- Engineered a **Knight's Tour** solution using **C++ and recursive backtracking** to find a valid sequence that visits every **chessboard square** exactly once.
- Employed **recursion, arrays, backtracking**, and **array-based board traversal** to explore valid knight moves and prune invalid states efficiently.
- Applied core **Data Structures & Algorithms** knowledge, along with **debugging and complexity analysis**, to generate valid **Knight's Tour paths** in **Node.js** and **JavaScript**.

SpeakCoach AI | Node.js, JavaScript, HTML, CSS | Live — Apr' 26

- Designed a **Node.js backend** for an AI-powered public speaking coach, exposing **REST API endpoints** for speech analysis, coaching chat, and practice generation.
- Integrated the **OpenRouter API** to power real-time speech analysis, AI coaching conversations, and personalized practice prompts.
- Delivered **token-based user authentication** with hashed passwords and deployed the application on **Render** for public access.

TRAINING

Basics Of DSA | Certificate — Jun' 26 – Jul' 26
Lovely Professional University

- Undertook intensive training in **Data Structures and Algorithms** using **C++**, covering **recursion, arrays, searching, sorting, linked lists, stacks, queues, trees**, and **algorithmic problem-solving**.
- Executed **recursion, backtracking, arrays, and state-space exploration** to develop and test a **Knight's Tour** project with **Node.js** and **JavaScript** integration.
- Enhanced **algorithmic thinking, debugging, complexity analysis, and problem-solving** skills through hands-on **DSA implementation and testing**.

CERTIFICATES

Database Management System Part1 | Infosys | Certificate — July' 26
Generative AI | Oracle | Certificate — Oct' 25
Data Science | Oracle | Certificate — Oct' 25
ReactJS | TechVeda | Certificate — Mar'25

ACHIEVEMENTS

ADOBE HACKATHON | CERTIFICATE — Aug' 26

- Qualified up to **Round 2** of the **Adobe Hackathon** through **team-based problem-solving, ideation**, and **technical solution development**.

CROWD COMPUTE PROJECT | CERTIFICATE — Apr' 25

- Earned a **₹25,380 stipend** through a **Google** Crowd Compute project focused on Prompt Engineering, Generative AI, AI model interaction, and prompt evaluation.

LEETCODE | PROFILE — Present

- Solved **50+** data structures and algorithms questions spanning fundamental, intermediate, and advanced level.

EDUCATION

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology — Aug' 24 – Present
Computer Science and Engineering; CGPA: 8.01

S.S.S.S.S Gurukulum — Dhanbad, Jharkhand
Intermediate (PCM) Percentage: 82% — Mar' 22 – May' 23

D.A.V Public School — Dhanbad, Jharkhand
Matriculation Percentage: 86.2% — Mar'20 – May' 21